PROBLEMS FOR THE 12^{th} INTERNATIONAL TOURNAMENT OF YOUNG MATHEMATICIANS

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Contents

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Notation

$\mathbb{N} = \{1, 2, 3, \ldots\}$	set of natural numbers (positive integers)
$\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$	ring of residue classes modulo n
$\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$	sets of integer, rational, real and complex numbers
$\mathbb{R}^n$	n-dimensional real space
$]a, b[$ and $[a, b]$	open and closed interval from a to b
$[\![a, b]\!]$	set of integers from a to b
$\mathbb{Z}[i]$	ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$
$\mathrm{diam}(G)$	diameter of a graph G
$\#M$ or $\|M\|$	cardinality of a set M
$M \times N$	Cartesian product of two sets or graphs M and N
$M \cup N, M \cap N, M \setminus N$	union, intersection and difference of two sets M and N
$M \triangle N$	symmetric difference of two sets M and N

1. Cutting into Pieces

Let n be a positive integer.

1. An $n \times n$ square is cut into n congruent pieces along the grid lines. (Two pieces are called *congruent* if one can be obtained from another by rotation or reflection.) What could the total length of the cuts be?

2. An $a \times b$ rectangle is cut into n congruent pieces along the grid lines, where n divides ab. What could the total length of the cuts be?

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3. An $n \times n$ square is cut along the grid lines into n pieces with the same area. What are the minimum and maximum total lengths of the cuts?

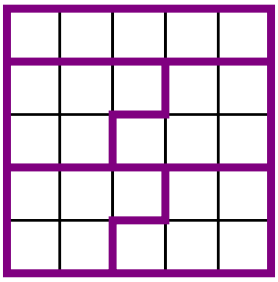

FIGURE 1. A 5×5 square is cut into 5 pieces with the same area, and the total length of the cuts is 16.

4. Let a and b be positive integers such that ab is divisible by n. An $a \times b$ rectangle is cut along the grid lines into n pieces with the same area. What are the minimum and maximum total lengths of the cuts?

5. Let P be a polygon in the real plane. Suppose that it is cut into n pieces with the same area so that every cut is parallel to a side of P (the side doesn't have to be the same for different cuts). Denote by $l(P, n)$ and $u(P, n)$ the minimum and maximum possible total lengths of the cuts respectively. Find or estimate the values of $l(P, n)$ and $u(P, n)$ when

a) P is a rectangle;
b) P is a regular polygon.

6. Finally, suppose that cuts are arbitrary straight lines (not necessarily parallel to sides). Denote by $L(P, n)$ and $U(P, n)$ the minimum and maximum possible total lengths of the cuts of a shape P into n pieces with the same area. Find or estimate the values of $L(P, n)$ and $U(P, n)$ when

a) P is a rectangle;
b) P is a regular polygon;
c) P is a circle;
d) P is another shape of your choice.

7. Suggest and investigate additional directions of research.

2. Coding Functions

We will say that a function $f : A \to B$ is *coding*, or that f *codes* the set A, if it is injective:

$$f(x) \neq f(y) \quad \text{for any distinct } x, y \in A.$$

Let n and $k \geq n$ be positive integers.

1. Find all $a, b \in \mathbb{Z}_n$ for which the linear function $f(x) = ax + b$ codes the set $\mathbb{Z}_n$.

2. Find all $a, b, c \in \mathbb{Z}_n$ for which the quadratic function $f(x) = ax^2 + bx + c$ codes $\mathbb{Z}_n$.

3. Give examples and find properties of polynomials of degree $d \geq 3$ which are coding functions from $\mathbb{Z}_n$ to $\mathbb{Z}_n$.

4. More generally, let $P(x) = a_d x^d + \cdots + a_1 x + a_0$ be a polynomial of degree d with integer coefficients. When are the values $P(0), P(1), \ldots, P(n-1)$ pairwise distinct modulo k?

5. Find necessary and/or sufficient conditions for a polynomial $P(x) = a_d x^d + \cdots + a_1 x + a_0$ with coefficients in $\mathbb{Z}_n$ to code the set $\mathbb{Z}_n[i] = \left\{ x + y\sqrt{-1} \mid x, y \in \mathbb{Z}_n \right\}$.

6. Let $\zeta \in \mathbb{C}$ be an *nth root of unity*, that is, a solution of the equation $x^n = 1$. Find necessary and/or sufficient conditions for a polynomial $P(x) = a_d x^d + \cdots + a_1 x + a_0$ with integer coefficients to code the set $\left\{\zeta, \zeta^2, \ldots, \zeta^n\right\}$. Start with $d = 1, 2, 3$.

7. Suggest and investigate further directions of research.

3. **Two Types of Functions**

Let f be a function defined and continuous on a interval $[a, b]$, where $a, b \in \mathbb{R}$ and $a < b$. We will say that f is a function of *first type* if, for all $x \in [a, b]$, the following inequality holds:

$$f(x) \leq \int_a^b f(s)ds.$$

We will say that f is a function of *second type* if, for all $x \in [a, b]$, one has:

$$f(x) \leq \int_a^x f(s)ds.$$

1. Find all $c \in \mathbb{R}$ for which the function $f(x) = x + c$ is a) of first type; b) of second type.

2. Find all $c \in \mathbb{R}$ for which the function $f(x) = cx - x^2$ is a) of first type; b) of second type.

3. Is it true that any function of second type is also of first type?

4. Give examples and find properties of functions of first type which are not of second type?

5. Find all real a and b for which $f(x) = \sin x$ is a function a) of first type; b) of second type.

6. Is it true that any differentiable function of second type is a solution of the following system of inequalities

$$\begin{cases} f'(x) \leq f(x), \text{ for all } x \in [a, b]; \\ f(a) \leq 0. \end{cases}$$

Is the converse true?

7. Determine the polynomials of degree $n \in \mathbb{N}$ which are functions a) of first type; b) of second type.

8. Find necessary and/or sufficient conditions for a function to be a) of first type; b) of second type.

9. Give examples of functions $f(x, y)$ defined and continuous on $[0, 1] \times [0, 1]$ such that, for all x, $y \in [0, 1]$, one has

a) $f(x, y) \leq \iint\limits_{[0,1]\times[0,1]} f(t, s)\mathrm{d}t\mathrm{d}s.$

b) $f(x, y) \leq \iint\limits_{[0,x]\times[0,y]} f(t, s)\mathrm{d}t\mathrm{d}s.$

10. Investigate an analogous problem for functions that are not necessary continuous but Riemann integrable, Lebesgue integrable, etc.

4. A Glass of Juice

A mathematician ordered a glass of juice. Let a and b be positive real numbers. Consider the following glasses (the wall of the glass in the first four cases is a solid of revolution of a curve $f(x,z)=0$ around the z-axis):

a) cylinder: $f(x,z)=x-a$ and $0\le z\le b$;
b) cone: $f(x,z)=z-ax$ and $0\le x\le b$;
c) hemisphere: $f(x,z)=z-\sqrt{a^2-x^2}$ and $0\le x\le a$;
d) parabola: $f(x,z)=z-x^2$ and $0\le x\le b$;
e) a right n-prism with height b and a regular n-gon with side a as its bases;
f) an upturned right n-pyramid with height b and a regular n-gon with side a as its base.

In the beginning, the glass was full of juice. Denote by V the volume of juice in it. While delivering the glass, the waiter took a misstep so that the glass made an angle α with its initial position. Hence, some juice was spilled (see Figure 2).

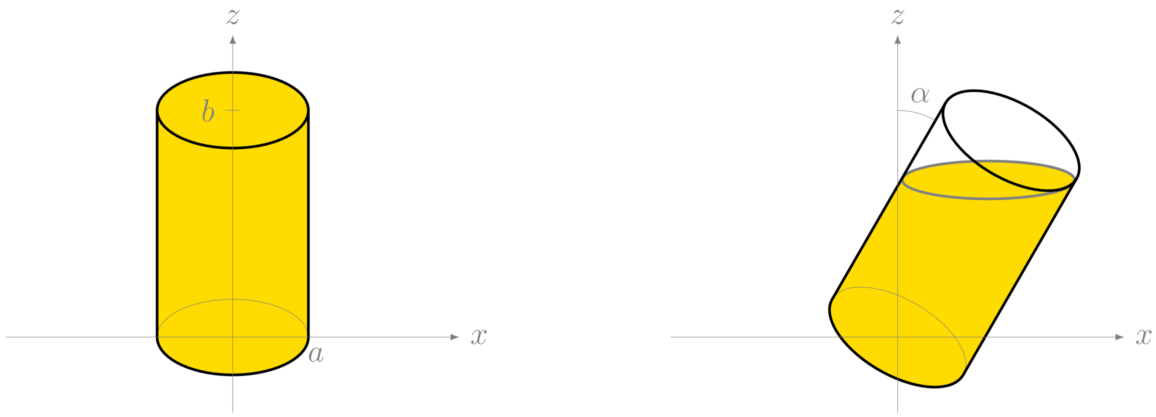

Figure 2. Juice was spilled from a cylinder glass.

1. For given a, b, n and a form of the glass above, does the volume of the remaining juice depend only on α?

Denote by $V(\alpha)$ the volume of the remaining juice. For the above glasses, answer the following questions.

2. Let $\alpha=30°$, $a=1$ and $b=3$. Find all possible values of $V(\alpha)/V$.

3. Let $V(\alpha)/V=1/2$, $a=1$ and $b=3$. Find all possible values of α.

4. Try to find a formula or all possible values of $V(\alpha)/V$ in terms of a, b, n and α.

5. Introduce other types of glasses and study the problem for them.

5. Multiplication Tables

Let n and $k\ge 2$ be positive integers. The *n-multiplication table of dimension k* is the set of all integers obtained as a product of k integers between 1 and n.

For a positive integer m, we define

$$a_k(m,n)=\#\left\{(x_1,x_2,\dots,x_k)\in[\![1,n]\!]^k \;\middle|\; \prod_{i=1}^{k}x_i=m\right\},$$

where $[\![1,n]\!]=\{1,\dots,n\}$.

1. Let α be a positive integer and p be a prime number, determine $a_k(p^\alpha,n)$

2. Find bounds for $a_k(m,n)$.

3. Find the following limits:

a) $\lim\limits_{n\to\infty} \frac{1}{n}\sum_{m=1}^{n} a_k(m,n)$;

b) $\lim\limits_{n\to\infty} \frac{1}{n^k}\sum_{m=n}^{n^k} a_k(m,n)$.

Denote by $M_k(n)$ the cardinality of the n-multiplication table of dimension k. For a sequence of integers $X = \{x_i\}_{i=0}^{\infty}$, denote by $M_k(X,n)$ the number of integers in X which appear in the n-multiplication table of dimension k.

4. Determine $\lim\limits_{n\to\infty} \frac{M_k(n)}{n^k}$.

5. Estimate (find bounds and an asymptotic) the values of $M_k(X,n)$ and $\lim\limits_{n\to\infty} \frac{M_k(X,n)}{\#(X\cap[\![1,n]\!])^k}$ in the following cases:

a) X is the sequence of prime numbers;
b) $X = \mathbb{N}$ and $k = 2$;
c) $X = \{i^r\}_{i\in\mathbb{N}}$ where r is a positive integer;
d) X is a geometric progression with $x_0 = 1$;
e) X is an arithmetic progression with the common difference being a prime number.

6. Find bounds and an asymptotic for $M_k(n)$.

7. Given k, m and the prime decomposition of m, determine time complexity for finding the minimum integer $n \geq 1$ such that $a_k(m,n) \geq 1$.

8. Suggest and study additional directions of research.

6. **Sequences from Rooted Trees**

Let T be a finite rooted tree with n leafs. Define the sequence $S(T) = (a_1, \dots, a_n)$ as follows.

- Choose a path from the root r to one of its leafs x_1, and define $a_1 = 0$.
- Choose a new leaf x_2, so that the the number of common edges in the path from r to x_1 and the path form r to x_2 is minimal, and define a_2 this number of common edges.
- Choose the leafs $x_3, x_4, \dots, x_n$ in the same way recursively: for $2 \leq i \leq n-1$, we take x_{i+1} such that the path from r to x_{i+1} minimises the total number of common edges (counted multiple times if necessary) with the path from r to x_1, the path from r to x_2, $\dots$, the path form r ot x_i.

For example, for the tree in Figure 3, we can have $x_1 = 2$, $x_2 = 4$, $x_3 = 1$, $x_4 = 5$, $x_5 = 3$, and its sequence is $a_1 = 0$, $a_2 = 0$, $a_3 = 1$, $a_4 = 2$, $a_5 = 2$.

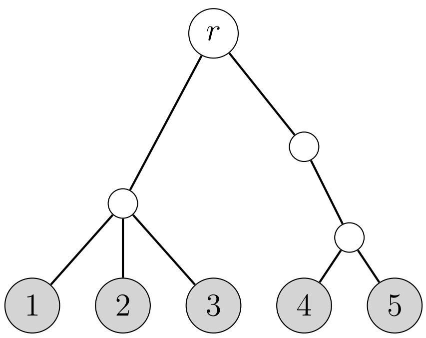

FIGURE 3. Example of a rooted tree.

1. Prove that the sequence $S(T)$ depends only on the tree T and not on the choice of leafs, that is $S(T)$ is well defined.

2. Calculate or describe the sequence $S(T)$ when T is

a) the tree in Figure 4;
b) a *full binary tree*, that is any non-leaf vertex has exactly two children;
c) a *full k-ary tree*, that is any non-leaf vertex has exactly k children;
d) a tree with all the vertices at the same distance h from the root having the same degree d_h, for a given sequence of positive integers $d_1, d_2, \ldots$.

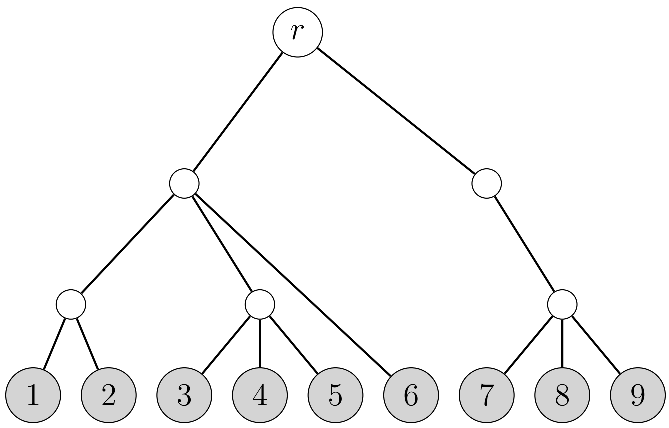

FIGURE 4. Another example of a rooted tree.

3. Is it true that if $S(T) = S(T')$, then T and T' are isomorphic?

4. Let $B = (b_1, b_2, \ldots)$ be a finite sequence of non-negative integers. Does the equation $S(T) = B$, where T is a finite tree, have a solution? Give necessary and sufficient conditions on B.

5. Estimate (give lower and upper bounds) and find properties of the last term a_n of the sequence $S(T)$, depending on the degrees of the vertices of T or other characteristics of T.

It is also possible to define the sequence $S(T)$ for a tree T of infinite height – instead of choosing leafs, one might consider infinite paths starting at the root.

6. Formulate and answer the previous questions for infinite trees. Also, in the question 3, investigate the case of a tree with each vertex having at least two descending edges.

7. Suggest and work on other directions of research.

7. **Families of Subsets**

Let m and n be positive integers, and let $\mathcal{F} = \{F_1, F_2, \ldots, F_m\}$ be a family of m subsets of the set $[\![1, n]\!] = \{1, \ldots, n\}$.

1. In this question suppose that m is odd.

a) Find the maximum possible value of m (in terms of n) for which there exists a family $\mathcal{F}$ with the property: for all $1 \le i \le m$, there are exactly $\frac{m-1}{2}$ indices $j \neq i$ such that $|F_i \cap F_j| > 0$.

b) Fix a positive integer k. Find the maximum possible value of m (in terms of n and k) for which there exists a family $\mathcal{F}$ with the property: for all $1 \le i \le m$, one has $|F_i| = k$ and there are exactly $\frac{m-1}{2}$ indices $j \neq i$ such that $|F_i \cap F_j| > 0$.

2. Given positive integers m, n and s, what maximum number of pairs (F_i, F_j) such that $|F_i \triangle F_j| = s$ could there be for a family $\mathcal{F}$? Start with $s = 1$. Here $A \triangle B = (A \setminus B) \cup (B \setminus A)$ denotes the *symmetric difference* of A and B.

3. Fix $\varepsilon \in]0, 1[$ and let n be sufficiently large.

a) Prove that there exist an integer k, depending on n and ε, and a family $\mathcal{F}$ with $m > (1 + 0.1\varepsilon^2)^n$ such that

$$|F_i| = k \quad \text{and} \quad |F_i \cap F_j| < \varepsilon k, \qquad \text{for all } 1 \leq i \neq j \leq m. \tag{1}$$

b) The same question for $m > (1 + 0.4\varepsilon^2)^n$.
c) Could you replace the constant 0.4 by a larger one?
d) Give an upper bound (in terms of n and ε) for m such that there exist a positive integer k and a family $\mathcal{F}$ satisfying (1).

4. Given n, find the maximum possible value of m for which there exists a family $\mathcal{F}$ with the following property:

a) $|F_i| \equiv a \pmod 2$ and $|F_i \cap F_j| \equiv b \pmod 2$ for all distinct $i, j \in [\![1, m]\!]$. Consider the cases $(a, b) = (0, 0), (0, 1), (1, 0)$ and $(1, 1)$ separately.

b) $|F_i| \equiv a \pmod 3$ and $|F_i \cap F_j| \equiv b \pmod 3$ for all distinct $i, j \in [\![1, m]\!]$. Consider the cases $(a, b) = (0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1)$ and $(2, 2)$ separately.

c) $|F_i| \not\equiv 0 \pmod 6$ and $|F_i \cap F_j| \equiv 0 \pmod 6$ for all distinct $i, j \in [\![1, m]\!]$.

5. Suggest and study additional directions of research.

8. **Primes with Special Properties**

For positive integers x and b, denote by $A_b(x)$ the product of the digits of x in its base b representation, and define $r_b(x)$ to be the integer obtained by writing the digits of x in its base b representation in the reverse order. For example, $A_{10}(1729) = 126$, $r_{10}(1729) = 9271$ and $A_3(1729) = 0$, $r_3(1729) = 1001012_3$.

Let $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, ... be the sequence of prime numbers. A prime p_n is called *b-product-like* if $A_b(p_n)$ has the same base b representation as n. A prime p_n is called *b-mirror-like* if $r_b(p_n) = p_{r_b(n)}$. For example, 73 is both 10-product-like and 10-mirror-like, since it is the 21st prime number, $7 \cdot 3 = 21$ and 37 is the 12th prime number.

1. a) Prove that 73 is the only prime which is both 10-product-like and 10-mirror-like.
b) Determine all primes which are both 9-product-like and 9-mirror-like.
c) Consider other bases b.

2. a) Classify all primes p_n which are 10-product-like.
b) The same question for any base b.

3. a) Are there infinitely many 10-mirror-like primes?
b) If yes, estimate the number of the 10-mirror-like primes not exceeding x, where $x \in \mathbb{N}$. If not, give an upper bound for the 10-mirror-like primes and try to classify them.

4. Given a positive integer ℓ, classify all primes p_n such that $|A_{10}(p_n) - n| = \ell$, if

a) $\ell = 1$;
b) $\ell > 1$.

5. Suggest and study additional directions of research.

9. **Convergence of Means**

Consider a sequence $x = \{x_n\}_{n=1}^{\infty}$ of positive real numbers. Denote $A_k(x,n) = \frac{x_{n+1}+\cdots+x_{n+k}}{k}$ and $G_k(x,n) = \sqrt[k]{x_{n+1}\cdot\ldots\cdot x_{n+k}}$. We will study the sequences

$$A_k(x) = \{A_k(x,n)\}_{n=1}^{\infty} \quad \text{and} \quad G_k(x) = \{G_k(x,n)\}_{n=1}^{\infty}.$$

1. Assume that the sequence $A_2(x)$ has a limit. Is it true that:

a) the sequence $\{x_n\}_{n=1}^{\infty}$ also has a limit?
b) the sequence $A_3(x)$ has a limit? If yes, may it be different from the limit of $A_2(x)$?
c) the sequence $G_2(x)$ has a limit? If yes, may it be different from the limit of $A_2(x)$?

2. Suppose that for some fixed $k \geq 3$ the sequence $A_k(x)$ has a limit $a > 0$.

a) Does this imply convergence of sequences $A_l(x)$, $G_l(x)$ for various $l \geq 1$? If they converge, what could be the possible values of their limits?
b) Answer the same question for sequences $A_\infty(x) = \{A_n(x,n)\}_{n=1}^{\infty}$ and $G_\infty(x) = \{G_n(x,n)\}_{n=1}^{\infty}$. Is it possible that a is not a *limiting point* for $\{x_n\}_{n=1}^{\infty}$ (that is $\{x_n\}_{n=1}^{\infty}$ does not contain subsequences converging to a)?
c) What are possible limiting points of the sequences in a) and b)?
d) Assume that for some k and l greater than 1 or equal to ∞ the sequences $A_k(x)$ and $G_l(x)$ converge to the same limit $a > 0$. Does this imply that the sequence $\{x_n\}_{n=1}^{\infty}$ also converges to the same limit?

3. Let $f\colon]0,+\infty[\to]0,+\infty[$ be an arbitrary function. Let R and T stand for any of the symbols A_k and G_l. For any sequence $x = \{x_n\}_{n=1}^{\infty}$ we shall assume that $A_1(x) = G_1(x) = x$. Finally, by $f(x)$ we denote the new sequence $\{f(x_n)\}_{n=1}^{\infty}$.

We say that the function f is (R,T)-*continuous*, if for any sequence x of positive real numbers and any $a > 0$ the condition $\lim_{n\to\infty} R(x) = a$ implies $\lim_{n\to\infty} T(f(x)) = f(a)$. Note that for $R = T = A_1$ this definition coincides with the standard definition of continuous functions.

a) Check which elementary functions (polynomials, trigonometric polynomials, $\exp(x)$, x^α) are (R,T) continuous for various R and T.
b) Do the theorems of Bolzano[1] and Weierstrass[2] generalise to (R,T)-continuous functions?
c) Is composition of two (R,T)-continuous functions also (R,T)-continuous?
d) Can an (R,T)-continuous non-constant function be periodic?
e) Investigate which other statements on continuous functions generalise or do not generalise to (R,T)-continuous functions?
f) Investigate the relationship between classes of (R,T)-continuous functions for various R and T.
g) Determine the cardinality (cardinal number) of the set of (R,T)-continuous functions.

4. Let $a(x,y) = \frac{x+y}{2}$ and $g(x,y) = \sqrt{xy}$ be the standard arithmetic and geometric means for $x, y > 0$. Let $0 < p < q$ be two fixed real numbers. Solve the functional equations

$$f(a(x,p)) + f(a(x,q)) = f(x) \tag{A}$$

$$f(g(x,p)) + f(g(x,q)) = f(x) \tag{B}$$

by finding all functions f defined on $[p,q]$, satisfying (A) and/or (B), which are

a) continuous;
b) periodic;
c) differentiable at least one time;
d) differentiable at least two times.

[1]**Bolzano theorem.** If a continuous function defined on an interval takes both positive and negative values, then it must be 0 at a point.

[2]**Weierstrass theorem**. A continuous function on a closed and bounded interval attains its extreme values.

5. Consider other modifications of the equations (A) and (B).

10. **An Electrical Network Problem**

Let G be an undirected connected graph without loops and multiple edges. For any two vertices u and v of the graph, denote by $d(u, v)$ the shortest path distance between u and v. The *diameter* of the graph, denoted by $\mathrm{diam}(G)$, is the maximal value of $d(u, v)$ over all pairs of vertices u and v. One can consider an electrical circuit associated with the graph G and define the effective resistance R as follows. Suppose that each edge corresponds to a 1 Ohm resistor, and connect a unit current source from u to v (see Figure 5). The *effective resistance* $R(u, v)$ is then the difference of the potentials at u and v.

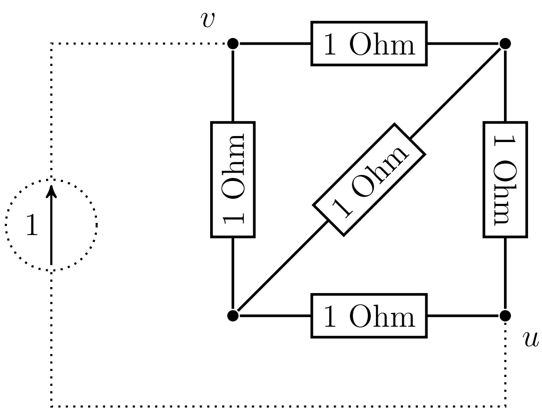

FIGURE 5. Connecting a unit current source between the vertices u and v.

1. Let V_G be the set of vertices of the graph G. A function $\rho : V_G \times V_G \to [0, +\infty[$ is called a *metric* on G if the following properties hold:

i) $\rho(u, v) = 0$ if and only if $u = v$;
ii) $\rho(u, v) = \rho(v, u)$ for any $u, v \in V_G$ (symmetry);
iii) $\rho(u, v) \le \rho(u, w) + \rho(w, v)$ for any $u, v, w \in V_G$ (triangle inequality).

Prove that the effective resistance R is a metric on G.

2. Is it true that $R(u, v)$ is maximal for some vertices u and v with $d(u, v) = \mathrm{diam}(G)$? Investigate this question for the following graphs.

a) $G = P_n \times P_n$, where P_n is a path on n vertices ('$\times$' denotes Cartesian product of graphs).
b) $G = C_n \times C_n$, where C_n is a cycle on n vertices.
c) $G = P_2^n$ is the graph of an n-dimensional hypercube.
d) $G = A_n$ is the graph of the permutohedron of order n. The *permutohedron* A_n is a polytope in $\mathbb{R}^n$, the vertices of which are formed by permuting the coordinates of the vector $(1, 2, \ldots, n)$, and two vertices $(a_1, \ldots, a_n)$ and $(b_1, \ldots, b_n)$ are connected if and only if one is obtained form the other by interchanging the numbers i and $i+1$ for some $1 \le i \le n-1$.
e) G is an arbitrary undirected connected graph.

3. For the above graphs, determine whether $R(u, v)$ attains its minimum for some vertices u and v such that $d(u, v) = 1$.

4. We say that the effective resistance is *monotonous* with respect to the shortest path distance if the inequality $d(u, v) < d(u', v')$ implies the inequality $R(u, v) \le R(u', v')$, for any pairs of vertices (u, v) and (u', v'). Investigate this monotonicity property for the above graphs.

5. Suggest and investigate additional directions of research.

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